

Erik Hogset

Mechanical Engineer | Sustainable Design, Energy Systems, and Manufacturing

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Portfolio: erikhogset.github.io

Education

Boston University College of Engineering, Boston, MA

Graduated May 2026

B.S. in Mechanical Engineering

Relevant Coursework: Heat Transfer • Fluid Mechanics • Mechanics of Materials • Electromechanical Design • Automation & Manufacturing Methods • Additive Manufacturing • Introduction to Clean Energy

Technical Skills

Manufacturing: CNC Milling & Turning, Manual Milling & Turning, 3D Printing (FFF, SLA), TIG Welding, CO₂ Laser Cutter, Waterjet Cutting, Sheet Metal Fabrication

CAD / CAM: SolidWorks, Autodesk Fusion 360, Onshape, CAM Programming & Toolpath Generation

Engineering Software: COMSOL Multiphysics, ANSYS Granta EduPack, LabVIEW, DAQ, Signal Processing

Programming / Controls: MATLAB, Python, C, G-code

Work Experience

Engineering Product Innovation Center (E.P.I.C.), Boston, MA.

May 2024 – May 2026

Laboratory Assistant

Supported prototyping and fabrication for student and faculty engineering projects in a high-traffic makerspace

Operated CNC mills, CNC lathes, manual mills/lathes, and 3D printers (*FFF* and *SLA*)

Generated CAM toolpaths and G-code for CNC machining using SolidWorks and Autodesk Fusion 360

Assisted with part fixturing, tolerancing, and design adjustments for manufacturability

Enforced lab safety and proper machine operation while supporting multiple users

Starbucks, Breezewood, PA and Boston, MA

Jun. 2021 – Sep. 2025

Barista

Maintained part-time employment while completing a full-time engineering course load

Worked in a fast-paced, team-based environment requiring accuracy, multitasking, and efficiency

Projects

Reactive Road Sign System – Senior Design I & II (*ME460/461*)

- Designed a retrofit mechanical system for roadside signposts to deflect under extreme wind loads
- Evaluated joint concepts using CAD, material selection, and structural failure analysis
- Modeled wind-induced loading and structural response to size a sacrificial breakaway pin
- Built and tested physical prototypes to validate loading behavior and manufacturability

Compact Fishing Kit Product – Automation & Manufacturing Methods (*ME345*)

- Designed a consumer product and developed a CAD-to-CAM workflow for CNC manufacturing
- Created CNC toolpaths and programmed collaborative robots for machine tending and assembly
- Integrated conveyors, robot programs, and CNC machines into an automated manufacturing cell

Sustainable Blender Redesign – Green Manufacturing (*ME535*)

- Conducted lifecycle eco-audit and redesign analysis of a countertop blender using ANSYS Granta EduPack
- Identified use-phase energy consumption as the dominant lifecycle environmental impact
- Evaluated motor redesign and recycled-material substitutions to reduce lifetime CO₂ and energy use

Electromechanical Interactive Ouija Board – Electromechanical Design (*ME360*)

- Built a 2-axis, belt-driven motion system to position a planchette using stepper motors
- Implemented motion control to convert AI-generated responses into coordinate targets

Adaptive LED Lamp – Introduction to Engineering Design (*EK210*)

- Designed and built a motion-activated LED lamp using electronic sensing and physical prototyping
- Implemented ambient-light sensing and time-based control to reduce blue-light exposure